



Efficacy of herbicides and fire to improve vegetative conditions for northern bobwhites in mature pine forests

by Judy D. J. Jones and Michael J. Chamberlain

Abstract Declining northern bobwhite (*Colinus virginianus*) populations during the past 30 years have prompted managers to seek ways to improve habitat quality for this species. Reductions in frequency of prescribed fire throughout considerable expanses of mature pine (*Pinus* spp.) forests have resulted in closed-canopy conditions, predominantly woody understory conditions, and a loss of early-successional habitats needed by bobwhites. Herbicides, particularly in conjunction with prescribed fire, may be useful for managing these pine forests to benefit the bobwhite and other early-successional species, but effects of herbicides in combination with fire are not well understood. Therefore, we used 3 similar-aged, mature pine stands to evaluate vegetative response to selective herbicides with prescribed fire with respect to bobwhite nesting and brood-rearing habitats. Our treatments were imazapyr with fire, imazapyr combined with glyphosate with fire, and dormant-season prescribed fire only. Plant diversity tended to decline on herbicide treatments during the first year but increased substantially on imazapyr plots during the second growing season following a burn. Bobwhite food plants increased following application of imazapyr during the first growing season and were greater for both herbicide treatments than burning alone during the second growing season. Abundance of hardwoods declined on both herbicide treatments. However, no treatments produced bare-ground percentages known to be selected by bobwhites, and only prescribed fire alone created and maintained suitable escape cover. Overall, imazapyr with fire provided the greatest net improvement in vegetative conditions for bobwhites and retained floristic diversity. We recommend that managers target areas in which vegetative conditions have progressed to where burning alone is incapable of restoring early-successional plant communities needed by the bobwhite and other species and apply imazapyr with fire to create diverse, patchy habitat for bobwhites.

Key Words *Colinus virginianus*, glyphosate, forest management, habitat management, herbicide, imazapyr, Louisiana, northern bobwhite, pine forest, prescribed fire

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Northern bobwhite (*Colinus virginianus*, hereafter bobwhite) populations have declined over the past 30 years throughout their range (Brennan 1991, Sauer et al. 1997). Bobwhites require a diverse, patchy habitat that includes open areas of herbaceous vegetation, a rich source of associated invertebrates, grassy areas for nesting, heavy brush or woody cover, and bare ground with little litter cover (Stoddard 1931, Stoddard 1962, Roseberry and Sudkamp 1998). It is postulated that bobwhite declines have resulted from losses of early-successional habitat (Thompson and DeGraaf 2001), changes in landscape use from small farms with brushy fence lines and hedgerows to clean-farming, silvicultural practices that create large monocultural stands, and, more recently, reduction of fire as a silvicultural tool (Klimstra 1982, Brennan 1991).

In the southeastern United States, bobwhites historically were associated with vast pine (*Pinus* spp.) forests maintained via frequent (every 3–5 years) fire. Although prescribed burning has been a traditional silvicultural practice, funding limitations, lack of qualified professionals and technicians, narrow windows for conducting

Therefore, our objective was to examine effects of 2 herbicide treatments in conjunction with prescribed fire and prescribed fire alone on vegetative structure within mature pine forests. Secondly, we evaluated efficacy of using selective herbicides with fire to renovate southern pine forests for bobwhites.

Study area

We conducted our research on the 700-ha Idlewild Research Station, part of the Louisiana State University Agricultural Center in East Feliciana Parish, Louisiana, USA. We conducted our experiment on 3 10.12-ha, 75–85-year-old, overmature, uneven-aged pine stands on hilltops and sloping terrain. All sites had been prescribed-burned on a 3–5-year rotation during the 20 years prior to our study. Each site was logged during August–December 1977 to reduce basal area toward 60–70 m²/ha (Langston 1981). During our study, basal area across sites averaged 30 m²/ha for pine and 40 m²/ha for hardwood.

These sites consisted mainly of loblolly (*P. taeda*) and shortleaf pine (*P. echinata*), and midstory hardwood growth of southern red oak (*Quercus falcata*), sweetgum (*Liquidambar styraciflua*), flowering dogwood (*Cornus florida*), water oak (*Q. nigra*), yaupon (*Ilex vomitoria*), winged elm (*Ulmus alata*), and black-

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burns, and risk have greatly reduced this practice throughout the Southeast (Haines et al. 2001). Reductions in fire and other disturbance ultimately will cause early-successional pine–grasslands to revert to thick hardwood midstory, which eventually becomes a closed forest canopy with little herbaceous vegetation (Engstrom et al. 1996).

Currently, wildlife managers can use prescribed fire only when conditions allow, often resulting in extended burning rotations that make it impossible to re-establish the native pine savannah community on many forested sites. Notably, herbicide applications have been used to improve habitat conditions for various wildlife species, including restoration of early-successional communities required by the bobwhite (Guynn et al. 2004, Miller and Miller 2004, Welch et al. 2004). However, little research has examined habitat quality for bobwhites within mature southeastern pine forests following application of selective herbicides (see Welch et al. 2004 in this section). Furthermore, it is important to understand how selective herbicides in conjunction with prescribed fire affect vegetative structure and, in turn, species such as the bobwhite.

gum (*Nyssa sylvatica*). Understory species consisted mainly of sweetleaf (*Symplocos tinctoria*), blackberry (*Rubus* spp.), waxmyrtle (*Myrica cerifera*), sparkleberry (*Vaccinium arboreum*), greenbriars (*Smilax* spp.), wild azalea (*Rhododendron* spp.), and Japanese climbing fern (*Lygodium japonicum*). The 3 stands were separated by even-aged pine stands and were a minimum of 0.32 km apart. Soils were mainly Providence silt loam (Type Fragiudalf) with 0–8% slopes on ridge tops (Langston 1981).

Drought conditions for Louisiana began in spring 1998 and continued through initiation of our study (National Climatic Data Center 2002). May through October 2000 was the driest period for Louisiana in the last 106 years. Average 2000 summer temperature and precipitation were 28°C and 25.55 cm, respectively. Cumulative effects for this period matched the worst drought occurrences of the twentieth century. Precipitation was near to above normal for summer 2001 (58.50 cm) and near normal for summer 2002 (39.24 cm). Summer temperature was near normal for both 2001 (27.1°C) and 2002 (27.3°C).

Methods

We applied an initial dormant-season burn in January and February 2000 to provide uniformity within stands before treatment application (Walstad and Kuch 1987). We burned one stand on 27 January 2000, and burn conditions were as follows: air temperature 8.33°C, surface winds E 29km/hour, and relative humidity 54% (National Weather Service 2002). We burned the remaining 2 stands on 17 February 2000, and burn conditions were as follows: air temperature 26.1°C, surface winds SE 16km/hour, and relative humidity 54% (National Weather Service 2002). We subdivided each 10.12-ha stand into 3 3.37-ha plots and used a bulldozer to establish plot perimeters. Control plots were those burned in 2000 and subsequently maintained on a 2-year burn rotation. We randomly assigned treatments ($n=2$, see below) to plots ($n=3$) within each of the 3 experimental stands.

We applied imazapyr (Arsenal AC®, BASF Corp., Research Triangle Park, N.C.) and imazapyr with glyphosate to treatment plots at rates of 226 g of imazapyr/0.405 ha (16 oz formulated product/acre) and 1,077.30 g of glyphosate+imazapyr/0.405 ha (38 oz formulated product/acre) in September 2000 for optimum timing of herbicide application (Miller 1989). We prepared herbicide mixtures immediately prior to application to minimize hydrolysis and degradation of herbicide in the spray tank. Imazapyr was used to control hardwoods and release herbaceous vegetation, and imazapyr was mixed with glyphosate to control imazapyr-tolerant species (e.g., blackberry). We broadcast-sprayed herbicides with a t-boom system mounted on a skidder. We applied imazapyr to appropriate treatment plots first, and then added glyphosate to the remaining imazapyr mixture and applied it to the imazapyr–glyphosate treatment plots.

We conducted a second-year post-treatment dormant-season burn in all plots during February 2002 to remove excess debris from experimental plots and for a 2-year burn rotation for control plots. We burned one stand on 14 February 2002, and burn conditions were: air temperature 18.3°C, surface winds NE 11km/hour, and relative humidity 25% (National Weather Service 2002). We burned 2 stands on 21 February 2002, and burn conditions were: air temperature 7.2°C, surface winds N 14km/hour, and relative humidity 75% (National Weather Service 2002).

All vegetation sampling occurred after the 2000 burn but before herbicide treatments and 2-years-post-herbicide treatments. We systematically located 5 circular, 0.04-ha subplots within each plot and divided each subplot into north, east, south, and west quadrants. We

measured vegetation composition, overstory canopy closure, vertical obstruction, average vegetation height (cm), maximum vegetation height (cm), and litter depth (cm) at subplot center and 10 m in each cardinal direction. We measured vegetation composition with a 0.5-m² Daubenmire frame to determine percentage coverage of grass, forb, woody, vine, debris, bare ground, and fern (Daubenmire 1959). We measured canopy closure with a forest densiometer (Lemmon 1956). We measured vegetation height with a 1.4-m Robel pole to determine vertical obstruction, and average and maximum vegetation heights (Greenfield et al. 2001). We used a 0.5-m pole with 0.1-cm increments to measure litter depth. We measured pine and hardwood species >10-cm dbh using a diameter tape and used an ultrasonic digital instrument at 1.4 m above ground to determine pine and hardwood composition within each subplot (Avery 1975, Wenger 1984). We determined an absolute count of number of stems <10-cm diameter within each subplot to determine midstory stem density. We used the line intercept method (Canfield 1941) on a 20-m diagonal through subplot center to determine plant species diversity and provide a measure of floristics 0.5 m above ground. These data are presented as proportion of the measured understory composed of each species. We identified plants to species for important bobwhite food plants and identified other plants to genus (Martin et al. 1951, Radford et al. 1968, Duncan and Duncan 1999, Miller and Miller 1999).

We used a randomized complete block analysis of variance to test the null hypothesis that there was no difference in mean plant response to herbicide treatments in conjunction with fire and prescribed fire alone. We blocked on stand, giving us 3 replicates of 3 treatments (prescribed fire, imazapyr with fire, imazapyr–glyphosate with fire) within 3 blocks. We constructed annual models using the SAS system for windows (SAS Institute, Inc. 1992) because the burn in 2002 was expected to create yearly differences in treatment and to evaluate treatment effects by year. Our first model tested response of understory structural characteristics, and a second model evaluated floristics and canopy coverage of preferred bobwhite foods relative to herbicide application and fire. To determine total proportion of bobwhite food per treatment, we created a new variable for this model called “food” that included plant genera and species important to bobwhite during summer months (3-seeded mercury [*Acalypha* spp.], ragweed [*Ambrosia* spp.], broomsedges [*Andropogon* spp.], spurred butterfly pea [*Centrosema virginianum*], partridge pea [*Chamaecrista fasciculata*], uniola and spranglegrasses [*Chasmanthium* spp.], butterfly pea [*Clitoria mariana*], panicgrasses [*Dichanthelium*

spp.], poorjoe [*Diodia teres*], lespedeza [*Lespedeza* spp.], honeysuckle [*Lonicera* spp.], nimblewill [*Muhlenbergia schreberi*], woodsorrel [*Oxalis* spp.], blackberry, horsenettle [*Solanum carolinense*], goldenrods [*Solidago* spp.], johnsongrass [*Sorghum* spp.], purpletop [*Tridens flavus*], blueberries and sparkleberries [*Vaccinium* spp.], and grape [*Vitis* spp.]). Past studies reported crop contents of bobwhites during the hunting season and rated importance of those species. However, we used only those research results that identified plant species important to bobwhites during summer months (nesting and brood-rearing). We used an *F*-test to determine treatment main effects and Tukey's Multiple Comparison to compare among treatments (Milliken and Johnson 1992). We used Shannon's Diversity Index (*H*) to evaluate plant diversity relative to treatment.

Results

Vegetation response

Prior to herbicide application, number of hardwoods >10 cm was greater ($F_{2,40}=11.37$, $P<0.001$) on plots subsequently treated with imazapyr (5.65 ± 0.64 ; Mean $\pm$ SE) and imazapyr–glyphosate (4.75 ± 0.39) than on control plots (2.66 ± 0.27), and litter depth was greater ($F_{2,40}=3.78$, $P=0.031$) on control (0.08 ± 0.02 cm) and plots subsequently treated with imazapyr–glyphosate (0.08 ± 0.02 cm) than on imazapyr (0.05 ± 0.01).

During the first growing season following application of herbicides, percentage of woody vegetation was greatest ($F_{2,40}=36.83$, $P<0.001$) in control plots, whereas debris was greatest ($F_{2,40}=54.78$, $P<0.001$) in both herbicide treatments (Table 1). Vertical obstruction ($F_{2,40}=45.92$, $P<0.001$), maximum vegetation height ($F_{2,40}=19.63$, $P<0.001$), average vegetation height ($F_{2,40}=18.86$, $P<0.001$), and hardwood midstory ($F_{2,40}=3.05$, $P=0.06$) were greatest in control plots. Number of hardwoods >10-cm dbh was greater on imazapyr plots ($F_{2,40}=4.14$, $P=0.02$) than on other treatments. Percentage of vines in the understory was greater ($F_{2,40}=3.04$, $P=0.06$) on imazapyr and control plots.

During the second growing season following application of herbicides and a dormant-season fire, percentage of woody vegetation ($F_{2,40}=20.66$, $P<0.001$), fern ($F_{2,40}=6.72$, $P=0.003$), vertical obstruction ($F_{2,40}=2.92$, $P=0.07$), maximum vegetation height ($F_{2,40}=17.65$, $P<0.001$), and average vegetation height ($F_{2,40}=36.61$, $P<0.001$) were greatest in control plots. Percentage of vine ($F_{2,40}=7.47$, $P=0.002$) and number of hardwoods >10-cm dbh ($F_{2,40}=8.41$, $P=0.001$) were greatest in imazapyr plots. Percentage of debris ($F_{2,40}=8.74$, $P=0.001$) and bare ground ($F_{2,40}=6.33$, $P=0.004$) were greatest in

Table 1. Mean vegetation structural characteristics ($\pm$ SE) that differed among 2 herbicide treatments (imazapyr and imazapyr–glyphosate) and prescribed fire during the first and second growing seasons post-treatment in 75–85-year-old pine forests of central Louisiana, 2001–2002.

Variable (growing season)	Treatment		
	Control	Imazapyr	Imazapyr glyphosate
Percentage woody understory (1)	60.0 (4.9)	30.0 (3.6)	16.8 (1.9)
Percentage debris in understory (1)	6.2 (1.9)	41.7 (6.1)	61.9 (3.1)
Vertical obstruction in meters (1)	0.62 (0.06)	0.12 (0.04)	0.07 (0.03)
Maximum vegetation height in meters (1)	1.43 (0.01)	0.84 (0.11)	0.77 (0.14)
Average vegetation height in meters (1)	1.05 (0.06)	0.40 (0.07)	0.54 (0.14)
Frequency of hardwood midstory (1)	8.3 (2.5)	3.0 (1.6)	2.9 (1.0)
Frequency of hardwood overstory (1)	2.9 (0.3)	4.4 (0.5)	3.8 (0.4)
Percentage of vines in understory (1)	19.3 (3.3)	19.9 (2.4)	11.8 (2.3)
Percentage of woody vegetation (2)	43.1 (4.8)	21.3 (3.1)	9.9 (2.8)
Percentage of fern in understory (2)	15.7 (5.3)	3.8 (1.9)	0.6 (0.3)
Vertical obstruction in meters (2)	0.95 (0.5)	0.25 (0.03)	0.08 (0.03)
Maximum vegetation height in meters (2)	1.39 (0.02)	1.11 (0.05)	0.87 (0.09)
Average vegetation height in meters (2)	0.71 (0.04)	0.41 (0.04)	0.26 (0.04)
Percentage of vine in understory (2)	17.7 (3.96)	36.2 (5.3)	16.7 (2.6)
Frequency of hardwood overstory (2)	2.8 (0.31)	5.9 (0.71)	4.3 (0.44)
Percentage of debris in understory (2)	10.7 (2.5)	20.1 (5.8)	45.5 (5.4)
Percentage of bare ground in understory (2)	0.5 (0.18)	3.5 (1.37)	8.1 (2.41)
Litter depth in centimeters (2)	0.1 (<0.1)	<0.1 (<0.1)	0.01 (<0.01)

imazapyr–glyphosate plots. Litter depth was greatest ($F_{2,40}=3.12$, $P=0.06$) in control and imazapyr–glyphosate plots.

Prior to application of herbicides, frequency of sweetgum was greater on plots subsequently treated with imazapyr–glyphosate ($F_{2,40}=5.97$, $P=0.005$). Overall, 21 tree species were identified (2 pine, 19 hardwood). Frequency of sweetgum remained greatest on imazapyr–glyphosate plots during the first and second growing seasons following the application of both herbicide treatments ($P\leq 0.05$). Only one species of hardwood (persimmon [*Diospyros virginiana*]) disappeared

on plots treated with either herbicide.

We noted minor differences in floristics among our plots prior to treatment. Passionflower (*Passiflora lutea*; $F_{2,40}=3.53$, $P=0.04$; 0.05 ± 0.03) and elephant's foot (*Elephantopus tomentosus*) were greatest on plots subsequently treated with imazapyr–glyphosate ($F_{2,40}=3.65$, $P=0.04$; 2.48 ± 0.61). Waxmyrtle ($F_{2,40}=3.38$, $P=0.04$; 2.48 ± 0.53) and poison ivy and sumacs (*Rhus* spp.; $F_{2,40}=3.75$, $P=0.03$; 5.55 ± 1.31) were greatest on plots subsequently treated with imazapyr.

During the first growing season following herbicide application, we noted various changes in floristics. Both herbicide treatments reduced red maple (*Acer rubrum*; $F_{2,40}=4.20$, $P=0.02$; 1.15 ± 0.44), American beautyberry (*Callicarpa americana*; $F_{2,40}=12.43$, $P<0.001$; 8.87 ± 2.49), dogwood ($F_{2,40}=4.00$, $P=0.03$; 2.82 ± 1.28), persimmon ($F_{2,40}=3.37$, $P=0.04$; 0.43 ± 0.22), sweetgum ($F_{2,40}=15.45$, $P<0.001$; 6.73 ± 1.63), Japanese climbing fern ($F_{2,40}=3.88$, $P=0.03$; 3.12 ± 1.54), oaks ($F_{2,40}=4.59$, $P=0.02$; 4.35 ± 1.39), poison ivy ($F_{2,40}=7.00$, $P=0.003$; 1.67 ± 0.66), and grape ($F_{2,40}=14.53$, $P<0.001$; 6.88 ± 1.66) as these species were all greatest in control plots. Partridge berry (*Mitchella repens*; $F_{2,40}=5.61$, $P=0.007$; 0.58 ± 0.19) was greatest in imazapyr–glyphosate plots. Overall, bobwhite foods were greatest in imazapyr plots ($F_{2,40}=5.50$, $P=0.008$; 24.00 ± 4.35). Pine seedlings were greatest in both herbicide treatments ($F_{2,40}=13.17$, $P<0.001$; 2.68 ± 0.79 , imazapyr; 4.35 ± 1.88 , imazapyr–glyphosate).

We noted continued changes in floristics during the second growing season. Beautyberry ($F_{2,40}=2.89$, $P=0.07$; 9.27 ± 2.28), sweetgum ($F_{2,40}=7.27$, $P=0.002$; 5.37 ± 1.77), oaks ($F_{2,40}=3.60$, $P=0.04$; 4.63 ± 1.84), sassafras (*Sassafras albidum*; $F_{2,40}=3.10$, $P=0.06$; 0.78 ± 0.47), waxmyrtle ($F_{2,40}=7.61$, $P=0.002$; 1.90 ± 0.46), blueberries ($F_{2,40}=2.91$, $P=0.07$; 1.25 ± 0.58), and grape ($F_{2,40}=4.56$, $P=0.02$; 7.03 ± 1.91) were greatest in control plots. Blueberries and sassafras were absent from herbicide plots. Elephant's foot ($F_{2,40}=3.57$, $P=0.04$; 2.08 ± 0.63) was greatest in imazapyr–glyphosate plots. Panicgrasses ($F_{2,40}=9.54$, $P<0.001$; 2.12 ± 0.48 , imazapyr; 6.35 ± 1.20 , imazapyr–glyphosate), burnweed (*Erechtites hieracifolia*; $F_{2,40}=4.57$, $P=0.02$; 0.82 ± 0.44 , imazapyr; 2.17 ± 0.81 , imazapyr–glyphosate), blackberry ($F_{2,40}=3.11$, $P=0.06$; 28.87 ± 4.07 , imazapyr; 15.02 ± 2.91 , imazapyr–glyphosate), and bobwhite food ($F_{2,40}=10.72$, $P<0.001$; 40.85 ± 4.40 , imazapyr; 30.72 ± 3.94 , imazapyr–glyphosate) were greatest in both herbicide treatments following the dormant-season fire prior to the second growing season. Poison ivy–sumac ($F_{2,40}=5.03$, $P=0.01$; 3.03 ± 0.94 , control; 0.07 ± 0.07 , imazapyr) was greatest in imazapyr and control plots. Diversity of plant species was similar across plots prior to treatment (Table 2). During the first growing season, diversity was qualita-

Table 2. Diversity of plant species (Shannon's Diversity Index) determined using line intercept prior to herbicide application (2000) and 2 years post-treatment (2001–2002) on 75–85-year-old pine stands in east-central Louisiana. All stands were burned in 2000 and 2002.

Year	Treatment		
	Control	Imazapyr	Imazapyr-glyphosate
2000	0.94	0.94	0.92
2001	0.95	0.72	0.58
2002	0.61	0.86	0.58

tively higher on control plots. However, plant communities were qualitatively more diverse on plots treated with imazapyr with fire during the second growing season.

Discussion

Both herbicide treatments in our study generally were more effective than prescribed fire alone at improving vegetation structure for brood-rearing and nesting bobwhites during the first year post-treatment. Both herbicides reduced percentage of woody vegetation close to the 37% described as habitat used by brood-rearing bobwhites in Mississippi (Taylor and Burger 2000) and the 20% described as good nesting habitat by Roseberry and Klimstra (1984). Additionally, hardwood midstory <10-cm dbh was approximately 2 times less on herbicide plots. Both herbicide treatments created more debris, but levels were similar to the 63–67% associated with greater numbers of brood-rearing bobwhites and the 67–71% preferred by nesting bobwhites in Missouri and Mississippi (Burger et al. 1994, Taylor and Burger 2000). Although debris and woody vegetation were at desirable levels for bobwhite nesting cover, herbaceous communities and perennial grasses typically associated with bobwhite brood-rearing and nesting habitat were not at desirable levels (Roseberry and Klimstra 1984, Tonkovich and Stauffer 1993, Burger et al. 1994, Taylor and Burger 2000). Control plots had greater vertical obstruction and came closest to habitat with >1-m vertical obstruction described by Davis (1964) as essential for bobwhite escape cover.

Plant species diversity tended to decline on both herbicide treatments during the first year post-treatment, as has been previously documented (see also Guynn et al. 2004, Miller and Miller 2004), but bobwhite food plants increased on imazapyr plots. During the first year post-treatment, Japanese climbing fern, brackenfern (*Pteridium aquilinum*), and grape dominated the understory on control plots, whereas pine seedlings were dominant on herbicide plots. Release of pine seedlings following application of glyphosate around larger pines was consistent with manufacturer specifications and other research studies

(Freedman et al. 1993, Glover and Zutter 1993, Lauer et al. 1993). Both herbicide treatments were effective at controlling undesirable species (hardwoods seedlings, Japanese climbing fern, brackenfern), but obviously killed or prevented release of some vegetation, thereby resulting in less species diversity than control plots initially.

During the second growing season, both herbicide treatments in conjunction with fire were more effective than prescribed fire alone at improving vegetational structure for brood-rearing and nesting bobwhites. The release of pine seedlings observed on both herbicide treatments during the first growing season was reduced substantially during the second growing season because of the dormant-season fire that occurred during 2002. Both herbicide treatments in conjunction with fire continued to reduce woody vegetation. Imazapyr with fire was more effective at controlling woody vegetation while reducing litter depth for nesting and brood-rearing bobwhites relative to imazapyr–glyphosate with fire, but all treatments were at desirable levels reported by Rosene (1969) for easy movement by bobwhites searching for food. Imazapyr application with fire increased vines (honeysuckle, jessamine [*Gelsemium* spp.], blackberry, greenbriar, and grape), and blackberry and grape were the dominant vine species encountered in our study. Blackberry is an important food source for bobwhites in summer, fall, and winter (Landers and Johnson 1976, McRae et al. 1979, Curtis et al. 1990).

All treatments produced low percentages of bare ground (<10%), yet bare ground is an essential component for quality bobwhite habitat because it improves foraging success of adults and chicks. It is plausible to assume that managers targeting mature pine forests similar to those we studied will have difficulty producing abundant bare ground because of needle cast and litter accumulation. Likewise, it is uncertain whether bobwhites foraging and rearing broods in forested systems require similar levels of bare ground as those in early-successional environments. That said, imazapyr–glyphosate with fire did increase bare ground similar to the 6% described by Burger et al. (1994) as being associated with optimal nesting habitat in Missouri. However, most studies have reported 25–75% bare ground as optimal nesting and brood-rearing habitat (Guthery 1986, Rice et al. 1993, Tonkovich and Stauffer 1993, Taylor and Burger 2000), and no treatments in our study fulfilled these requirements. If creating abundant bare ground is desirable in forests similar to those we studied, additional mechanical manipulations may be necessary.

Plant species diversity increased on plots treated with imazapyr in conjunction with fire during the second year post-treatment, and bobwhite food plants were greater on

both herbicide treatments with fire than on controls. Previous research has shown that herbicides generally reduce plant species diversity the first growing season after treatment, but plant communities generally recover or are more diverse in subsequent years (Guynn et al. 2004, Miller and Miller 2004). Likewise, although both herbicide treatments greatly reduced sweetgum, neither negatively affected hard-mast-producing species >10-cm dbh. Sweetgum are considered undesirable when managing pine forests for bobwhites, primarily because of their propensity to stump-sprout following disturbance (Wenger 1953) and their ability to outcompete desirable forbs and legumes. Conversely, mast-producing hardwoods are important for bobwhites, particularly during winter (Landers and Johnson 1976, McRae et al. 1979, Curtis et al. 1990). Our findings suggest that application of imazapyr and imazapyr with glyphosate likely will not affect availability of mature, mast-producing species, although regeneration will be negatively affected through reductions in woody understory species. However, long-term effects of imazapyr application on mast production are not known.

Vegetative response following herbicide application and fire in our study likely was negatively affected by relatively high overstory canopy coverage and extended drought conditions. Canopy-cover values ≤50% are typically associated with greater numbers of bobwhites (Stoddard 1931, Rosene 1969, Lee and Brennan 1994), and landscapes containing >50% overstory canopy coverage have been determined to be marginal or unsuitable (Stoddard 1931, Rosene 1969, Lee and Brennan 1994). Crowns of 75–85-year-old pines combined with scattered hardwood species on our plots produced greater canopy closure (i.e., 70–90%) than is desirable for bobwhites. Additionally, drought conditions that began in 1998 and continued through the initial stages of our study might have negatively affected growth of desired bobwhite vegetation. Effects of imazapyr and glyphosate herbicides may have been amplified by the drought conditions. Specifically, drought conditions can increase herbicide carryover because plants are not able to take up as much herbicide when they are moisture stressed, hydrolysis can not occur when soil is too dry, and microbial activity that breaks down herbicides almost stops (Parker 1993).

Management implications

Both herbicide treatments improved brood-rearing and nesting habitat for bobwhites. Overall, imazapyr with fire provided a greater benefit than imazapyr–glyphosate with fire and retained floristic species diversity. Herbicides typically are used to improve conditions for timber production, but they also are used to manage habitats for

wildlife (Guynn et al. 2004, Miller and Miller 2004). Herbicide use has increased in recent years because of restricted use of prescribed burning (Haines et al. 2001). However, prescribed burning historically has been used to manage and maintain pine and pine–hardwood forest ecosystems (Robbins and Myers 1992), and fire typically increases production of native legumes used by bobwhites (Lewis and Harshbarger 1986). Absence of disturbance in these forest ecosystems eventually will cause early-successional pine–grasslands to revert to thick hardwood midstory that eventually will become a closed pine–hardwood canopy with little herbaceous vegetation (Engstrom et al. 1996). Using mechanical disturbance (e.g., thinning) and herbicides can help prevent these closed-canopy conditions. On sites where prescribed fire can no longer create early-successional herbaceous communities, managers should consider using mechanical disturbance and herbicides to recapture sites, then prescribed fire to manage understory vegetation (Edwards et al. 2003).

Although application of imazapyr improved vegetative characteristics for northern bobwhites more than prescribed burning alone, our findings suggest that the greatest net benefit from imazapyr application occurred following the dormant-season fire prior to the second growing season. Managers should recognize potential beneficial effects of imazapyr application for bobwhites but also should be cognizant of the continued importance of prescribed fire in managing landscapes for the bobwhite, as demonstrated in our study.

Vegetative improvements from herbicide application and fire likely would have been greater (i.e., increased percentages of herbaceous vegetation) had canopy closure been reduced. Our stands had >70–90% canopy closure, but ideally <50% would be associated with higher-quality bobwhite habitat. It is likely that reduced canopy coverage on our research plots would have improved response of many herbaceous species following herbicide application and prescribed fire during the second growing season. Future research should evaluate vegetative response to herbicide application under variable canopy conditions.

Bobwhites require diverse, patchy habitat to successfully survive and reproduce; therefore, widespread application of imazapyr in forested stands would not achieve this goal. For instance, bobwhites prefer vegetation >1 m in height for escape cover. Prescribed fire alone created and maintained cover more suitable as escape cover for bobwhites in our study. We recommend that managers target areas where prescribed burns are not possible or areas where relative benefits of prescribed fire are low because of advanced understory succession (i.e., areas where stand renovation is necessary to achieve management objectives) and apply imazapyr in conjunction with

fire. Our findings suggest that application of imazapyr to these stands will improve vegetative characteristics for bobwhites and, coupled with the continuation of prescribed fire when possible, will help create diverse, patchy habitat typically associated with greater numbers of bobwhites. Alternatively, managers could apply herbicide in strips, leaving selected strips of vegetation unsprayed. This technique likely would maintain sufficient escape cover. In forests similar to those we studied, managers may need to use alternative mechanical methods (e.g., strip disking) following prescribed burning and herbicide application to produce bare ground for optimal nesting habitat and brood-rearing habitat.

Acknowledgments. We appreciate the field assistance provided by W. Wilson, K. Landry, D. Scognamillo, and L. Jezewski. We thank D. Sanders, G. Gentry, and W. Forbes for support throughout the study. D. Miller, W. Palmer, L. M. Conner, and T. Clason provided valuable comments on an earlier draft of the manuscript. Funding and support were provided by the Louisiana State University Agricultural Center through the McIntire-Stennis Forestry Program, BASF Corporation, and the School of Renewable Natural Resources at Louisiana State University.

Literature cited

- AVERY, T. E. 1975. Natural resources measurements. McGraw-Hill, New York, New York, USA.
- BRENNAN, L. A. 1991. How can we reverse the northern bobwhite population decline? *Wildlife Society Bulletin* 19: 544–555.
- BURGER, L. W., JR., M. R. RYAN, E. W. KURZEJESKI, AND T. V. DAILEY. 1994. Factors affecting the habitat value of conservation reserve program lands for northern bobwhite in northern Missouri. Pages 142–156 *in* M. Dicks and M. Monsoon, editors. Proceedings NC-163 post CRP land use conference. United States Department of Agriculture, Project 92110381.
- CANFIELD, R. H. 1941. Application of the line interception method in sampling range vegetation. *Journal of Forestry* 39: 388–394.
- CURTIS, P. D., T. L. SHARPE, AND P. D. DOERR. 1990. Early summer diet of male northern bobwhite in the North Carolina sandhills. *Proceedings of the Annual Conference of Southeastern Association of Fish and Wildlife Agencies* 44: 250–259.
- DAUBENMIRE, R. K. 1959. A canopy coverage method of vegetational analysis. *Northwest Science* 35: 43–64.
- DAVIS, C. A. 1964. Components of the habitat of bobwhite quail in Payne County, Oklahoma. Dissertation, Oklahoma State University, Stillwater, USA.
- DUNCAN, W. H., AND M. B. DUNCAN. 1999. Wildflowers of the eastern United States. University of Georgia Press, Athens, USA.
- EDWARDS, S. L., S. DEMARIS, B. WATKINS, AND B. K. STRICKLAND. 2003. White-tailed deer forage production in managed pine stands and summer food plots. *Southeast Deer Study Group Meeting* 26: 22–23.
- ENGSTROM, R. T., D. B. MCNAIR, L. A. BRENNAN, C. L. HARDY, AND L. W. BURGER. 1996. Influence of dormant versus lightning season prescribed fire on birds in longleaf pine forests: experimental design and preliminary results. *Transactions of the North American Wildlife and Natural Resources Conference* 61: 200–207.
- FREEDMAN, B., R. MORASH, AND D. MACKINNON. 1993. Short-term changes in vegetation after the silvicultural spraying of glyphosate herbicide

- onto regenerating clearcuts in Nova Scotia, Canada. *Canadian Journal of Forest Research* 23: 2300-2311.
- GLOVER, G. R., AND B. R. ZUTTER. 1993. Loblolly pine and mixed hardwood stand dynamics for 27 years following chemical, mechanical, and manual site preparation. *Canadian Journal of Forest Research* 23: 2126-2132.
- Greenfield, K. C., L. W. Burger, Jr., and M. J. Chamberlain. 2001. Herbicide and prescribed fire as habitat management tools for northern bobwhite in Conservation Reserve Program fields. *Proceedings of the Annual Conference of Southeastern Association of Fish and Wildlife Agencies* 55: 445-455.
- GUTHERY, F. S. 1986. Beef, brush, and bobwhites. Caesar Kleberg Wildlife Research Institute Press, Texas A&I University, Kingsville, USA.
- GUYNN, D. C., S. T. GUYNN, T. B. WIGLEY, AND D. A. MILLER. 2004. Herbicides and forest biodiversity—what do we know and where do we go from here? *Wildlife Society Bulletin* 32: 1085-1092.
- HAINES, T. K., R. L. BUSBY, AND D. A. CLEAVES. 2001. Prescribed burning in the south: trends, purpose, and barriers. *Southern Journal of Applied Forestry* 25: 149-153.
- KLIMSTRA, W. D. 1982. Bobwhite quail and changing land use. *Quail II*: 1-5.
- LANDERS, J. L., AND A. S. JOHNSON. 1976. Bobwhite quail food habits in the southeastern United States with a seed key to important foods. Tall Timbers Research Station Miscellaneous Publication Number 4.
- LANGSTON, H. W., JR. 1981. Effects of pine reproduction cutting systems on soils and understory vegetation in southeast Louisiana. Dissertation, Louisiana State University, Baton Rouge, USA.
- LAUER, D. K., G. R. GLOVER, AND D. J. GJERSTAD. 1993. Comparison of duration and method of herbaceous weed control on loblolly pine response through mid-rotation. *Canadian Journal of Forest Research* 23: 2116-2125.
- LEE, J. M., AND L. A. BRENNAN. 1994. Changes in northern bobwhite habitat and populations in a southern Mississippi wildlife management area: 1955-1992. *Proceedings of the Annual Conference of Southeastern Association of Fish and Wildlife Agencies* 48: 201-207.
- LEMMON, P. E. 1956. A spherical densiometer for estimating forest overstory density. *Forest Science* 1: 314-320.
- LEWIS, C. E., AND T. K. HARSHBARGER. 1986. Burning and grazing effects on bobwhite foods in the southeastern coastal plain. *Wildlife Society Bulletin* 14: 455-459.
- MARTIN, A. C., H. S. ZIM, AND A. L. NELSON. 1951. *American wildlife and plants: a guide to wildlife food habits*. Dover Publications, New York, New York, USA.
- MCRAE, W. A., J. L. LANDERS, J. L. BUCHNER, AND R. C. SIMPSON. 1979. Importance of habitat diversity in bobwhite management. *Proceedings of the Annual Conference of Southeastern Association of Fish and Wildlife Agencies* 33: 127-135.
- MILLER, J. H. 1989. Optimum timing for ground-applied forestry herbicides in the south. *United States Department of Agriculture Forest Service Management Bulletin* R8-MB28.
- MILLER, J. H., AND K. V. MILLER. 1999. *Forest plants of the southeast and their wildlife uses*. Southern Weed Science Society. Craftmaster Printers, Auburn, Alabama, USA.
- MILLER, K. V., AND J. H. MILLER. 2004. Forestry herbicide influences on biodiversity and wildlife habitat in southern forests. *Wildlife Society Bulletin* 32: 1049-1060.
- MILLIKEN, G. A., AND D. E. JOHNSON. 1992. *Analysis of messy data*. Chapman and Hall, New York, New York, USA.
- NATIONAL CLIMATIC DATA CENTER'S CLIMATE. 2002. National Oceanic and Atmospheric Administration webpage. Online at <http://twf.ncdc.noaa.gov/oa/climate> (accessed October 2002).
- NATIONAL WEATHER SERVICE. 2002. National Oceanic and Atmospheric Administration webpage. Online at <http://www.boi.noaa.gov/FIREWX/NEWFWFNEW.html> (accessed October 2002).
- PARKER, R. 1993. Drought Advisory EM4858. What happens to herbicides in the soil under drought conditions? Washington State University Cooperative Extension Service, Miscellaneous Publication, Pullman, USA.
- RADFORD, A. E., H. E. AHLES, AND C. R. BELL. 1968. *Manual of the vascular flora of the Carolinas*. University of North Carolina Press, Chapel Hill, USA.
- RICE, S. M., F. S. GUTHERY, G. S. SPEARS, S. J. DEMASO, AND B. H. KOERTH. 1993. A precipitation-habitat model for northern bobwhites on semiarid rangeland. *Journal of Wildlife Management* 57: 92-102.
- ROBBINS, L. E., AND R. L. MYERS. 1992. Seasonal effects of prescribed burning in Florida: a review. A Tall Timbers Research, Inc., Miscellaneous Publication No. 8., Tallahassee, Florida, USA.
- ROSEBERRY, J. L., AND W. D. KLIMSTRA. 1984. *Population ecology of the bobwhite*. Southern Illinois University Press, Carbondale, USA.
- ROSEBERRY, J. L., AND S. D. SUDKAMP. 1998. Assessing the suitability of landscapes for northern bobwhite. *Journal of Wildlife Management* 62: 895-902.
- ROSENE, W. 1969. *The bobwhite quail: its life and management*. Rutgers University Press, New Brunswick, New Jersey, USA.
- SAS INSTITUTE. 1992. *SAS/Stat user's guide, version 6*. Fourth edition. SAS Institute, Incorporated, Cary, North Carolina, USA.
- SAUER, J. R., S. SCHWARTZ, B. G. PETERJOHN, AND J. E. HINES. 1997. *The North American Breeding Bird Survey Home Page, version 94.3*. Patuxent Wildlife Research Center, Laurel, Maryland, USA.
- STODDARD, H. L. 1931. *The bobwhite quail: its habits, preservation, and increase*. Charles Scribner's Sons, New York, New York, USA.
- STODDARD, H. L. 1962. Some techniques of controlled burning in the deep southeast. *Tall Timbers Fire Ecology Conference Proceedings* 1: 133-144.
- TAYLOR, J. D., AND L. W. BURGER, JR. 2000. Habitat use of breeding bobwhite in managed oldfield habitats in Mississippi. *Quail IV*: 4: 7-15.
- THOMPSON, F. R., AND R. M. DEGRAAF. 2001. Conservation approaches for woody, early successional communities in the eastern United States. *Wildlife Society Bulletin* 29: 483-494.
- TONKOVICH, M. J., AND D. F. STAUFFER. 1993. Evaluating micro-habitat selection by northern bobwhite in Virginia. *Proceedings of the Annual Conference of Southeastern Association of Fish and Wildlife Agencies* 47: 257-267.
- WALSTAD, J. D., AND P. J. KUCH. 1987. *Forest vegetation management for conifer production*. John Wiley and Sons, New York, New York, USA.
- WELCH, J. R., K. V. MILLER, W. E. PALMER, AND T. B. HARRINGTON. 2004. Response of understory vegetation important to northern bobwhite following Imazapyr and mechanical treatments. *Wildlife Society Bulletin* 32: 1071-1076.
- WENGER, K. F. 1953. The sprouting of sweetgum in relation to season of cutting and carbohydrate content. *Plant Physiology* 28: 35-49.
- WENGER, K. F. 1984. *Forestry Handbook*. Second edition. John Wiley and Sons, New York, New York, USA.

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